Nonintrusive Cross-Motor Detection and Severity Assessment of False-Data-Injection Attacks in Heterogeneous Inverter-Fed Drives

Abolfazl Najar, He Yang, Jin Ye, *Senior Member, IEEE*, Ping Ma and WenZhan Song, *Senior Member, IEEE*

***Abstract*—This paper evaluates false-data-injection attack (FDIA) detection and injected-bias severity assessment in heterogeneous inverter-fed drives using trusted motor-terminal or point-of-common-coupling (PCC) measurements. Low-frequency, high-frequency, and time-domain features are combined with an InceptionTime classifier. Laboratory data comprise 18 recordings from an alternating current induction motor (ACIM) and a permanent magnet synchronous motor (PMSM), covering three speeds and three load settings. Training, validation, and testing use disjoint recordings and episodes, with three repeated holdouts. Mean binary detection accuracy is 87.41% for motor-side sensing and 80.78% for PCC sensing; corresponding four-class severity balanced accuracies are 90.81% and 55.62%. The proposed model attains the highest mean PCC balanced accuracy among the evaluated baselines, although it does not lead every severity metric. The PCC severity model identifies 61.39%, 85.08%, and 88.09% of low-, medium-, and high-bias attacks as attacks, respectively, with a 37.23% normal false-alarm rate. Supervised PMSM-to-ACIM fine-tuning reaches 91.38% PCC balanced accuracy at a nominal 20% target adaptation budget, while the reverse direction remains difficult. Removing high-frequency features reduces PCC processing time from 22.37 to 12.32 ms with similar detection balanced accuracy. These results characterize the capabilities and limitations of PCC monitoring on the tested platform under additive feedback-bias attacks.**

***Index Terms*—False-data-injection attack, point of common coupling, inverter-fed motor drives, heterogeneous motor drives, nonintrusive monitoring, cross-motor adaptation, severity assessment, cyber-physical security**

I. Introduction

FDIA detection and severity assessment are important in networked motor-drive systems, where manipulated sensor feedback can degrade control performance and cause unsafe operation [1]. Increasing drive connectivity makes timely detection of sensor-oriented cyber attacks essential for resilient operation [2], [3]. Current signals are attractive for FDIA monitoring because they are easy to acquire, strongly coupled to electromechanical dynamics, and have been shown to reveal sensor-integrity attacks in motor drives [4], [5]. However, most existing studies emphasize terminal-side sensing, operate on a single machine domain, or assume sufficient labeled data for supervised training. These assumptions limit practicality in heterogeneous deployments where different motor types, sensing locations, and data availability constraints all affect performance. Motivated by these limitations, this work experimentally evaluates FDIA detection in a heterogeneous motor-drive setting involving both ACIM and PMSM domains. The study evaluates binary detection (`fdia2`), four-level injected-bias assessment (`severity4`), cross-domain transfer, and processing cost for converter-fed industrial loads connected at a shared electrical interface. The framework combines low-frequency (LF), high-frequency (HF), and time-domain (TD) electrical features with an InceptionTime classifier, and compares motor-terminal and PCC sensing.

A. Challenges at PCC in Heterogeneous Multi-Motor Systems

From a deployment perspective, motor-terminal measurements provide cleaner machine-specific signatures, whereas PCC measurements are more scalable and non-intrusive. PCC sensing is attractive because it avoids direct motor-terminal instrumentation and is compatible with shared electrical infrastructure [6]. However, PCC currents are affected by converter harmonics, network interactions, and load aggregation. Consequently, PCC attack signatures are weaker and less separable than motor-terminal signatures [7]. The problem becomes more difficult in heterogeneous multi-motor settings. ACIM and PMSM systems exhibit different current characteristics even under the same attack framework, which introduces a substantial domain shift between source and target deployments. Consequently, a detector trained on one motor family may generalize poorly to the other unless explicit adaptation is used. Therefore, this study evaluates in-domain performance, source-only transfer, few-shot adaptation, and transfer-direction asymmetry.

B. Contributions

The main contributions of this work are as follows:

- **Evaluated** nonintrusive PCC-based FDIA detection on a heterogeneous ACIM/PMSM laboratory platform using whole-recording holdouts that also keep all constituent episodes disjoint.
- **Quantified** the distinction between exact injected-bias classification and detecting an attack of a given bias level, reporting classwise recall together with normal false alarms and repeated-holdout variability.
- **Compared** source-only transfer and supervised adaptation, and measured the accuracy–processing-cost tradeoff

Abolfazl Najar, He Yang, Jin Ye, and WenZhan Song are with the College of Engineering, University of Georgia, Athens, GA 30602, USA (e-mail: najar@uga.edu; heyang95@uga.edu; jin.ye@uga.edu; wsong@uga.edu).

Ping Ma is with the Department of Statistics, University of Georgia, Athens, GA 30602, USA (e-mail: pingma@uga.edu).

Corresponding author: Jin Ye (e-mail: jin.ye@uga.edu).

of LF/HF/TD representations and validation-ranked feature subsets.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III describes the physical system and dataset construction, while Section IV details the proposed FDIA detection framework and feature pipeline. Experimental results and cross-domain adaptations are reported in Section V. Section VI analyzes real-time deployment and latency, followed by a discussion of tradeoffs in Section VII. Section VIII concludes the paper.

II. Related Work

FDIA detection has been studied extensively in smart grids and broader cyber-physical systems (CPS), but the assumptions in that literature do not transfer cleanly to inverter-fed motor drives. Many grid- and CPS-oriented approaches rely on state estimation, observer-based detection, invariant analysis, adaptive neuro-fuzzy models, or graph-based learning [8], [9], [10], [11], [12], [13]. These studies demonstrate strong FDIA-detection capability at the system level, but they are generally not designed for drive-level current monitoring or for deployment at a shared point of common coupling (PCC).

Motor-drive-specific studies are more closely related to the present work. Prior work has shown that current signals can support effective FDIA detection in electric-drive systems and that transfer learning can improve performance when labeled data are limited [2], [3]. Related industrial-IoT monitoring studies have also considered cyber-detection in machine-monitoring settings [14]. However, these studies mainly rely on motor-terminal sensing, single-motor settings, or closely related operating domains, and therefore provide limited evidence for heterogeneous cross-motor transfer.

The main gap is not FDIA detection in general, which has been widely studied in smart grids and cyber-physical systems, but nonintrusive PCC-based monitoring for heterogeneous inverter-fed motor-drive systems. In particular, the existing literature does not clearly establish: 1) how PCC sensing compares with motor-side sensing for low-severity current-feedback FDIAs, 2) whether PCC measurements support reliable multi-level severity assessment rather than only binary detection, given that attack signatures become weaker and less separable under signal aggregation and harmonic distortion, and 3) how detection performance transfers between heterogeneous ACIM and PMSM domains when labeled target-domain data are limited. The contribution is a comparative experimental assessment rather than a new neural-network architecture. This work addresses these three questions experimentally on a shared ACIM/PMSM platform using multi-scale features, in-domain evaluation, source-only cross-domain transfer, and few-shot target-domain adaptation.

III. System and Data

A. Experimental Platform: ACIM/PMSM and PCC Sensing

Fig. 1 illustrates the overall setup, including the shared PCC, the two drive branches, the ACIM and PMSM loads, the sensing locations, and the FDIA injection points in the control/feedback paths. The experimental platform consists of a heterogeneous two-motor industrial setup including one ACIM branch and one PMSM branch. Both branches are supplied from a common upstream electrical interface, enabling measurements at the PCC in addition to motor-side measurements. This setup enables comparison of motor-terminal and PCC sensing under heterogeneous ACIM/PMSM dynamics. The dataset contains 540 scheduled attack episodes (270 for each motor), formed by repeating 30 bias patterns over nine speed/load combinations per motor. These episodes are not 540 independent motor installations. The experiments cover three speed operating points, namely 900, 1100, and 1300 rpm.

Two measurement streams were used throughout this study. The first stream corresponds to motor-side measurements, referred to as *motor* in the remainder of the paper. The second stream corresponds to PCC-side measurements, referred to as *PCC*. The PCC stream is the primary deployment-oriented sensing location because it avoids dedicated sensors at each machine. The measured signals and channel counts associated with the motor-side and PCC-side streams are summarized in Table II.

B. Attack Model, Sensing Assumptions, and Labels

In the testbed, the FDIA is injected into the controller's phase-current feedback path. The PCC and motor-side monitoring sensors are used as trusted measurements to record the physical response of the drive system under attack. The monitoring sensors are isolated from the compromised controller feedback path. Thus, PCC sensing represents the nonintrusive monitoring case, while motor-side sensing serves as a performance reference. Replay, denial-of-service, monitoring-sensor attacks, and malicious command manipulation are outside the scope of this study.

Let $\mathbf{i}_{abc}(t) = [i_a(t), i_b(t), i_c(t)]^\top$ denote the true three-phase current-feedback vector in the controller's per-unit representation, and let $\tilde{\mathbf{i}}_{abc}(t)$ denote the corrupted current-feedback vector received by the motor controller. During an attack interval, the FDIA is modeled as an additive phase-current bias:

$$\tilde{\mathbf{i}}_{abc}(t) = \mathbf{i}_{abc}(t) + \boldsymbol{\alpha}, \qquad t \in [t_{\mathrm{on}}, t_{\mathrm{off}}], \tag{1}$$

Here, $\boldsymbol{\alpha} = [\alpha_a, \alpha_b, \alpha_c]^\top$ denotes the injected per-unit bias vector, with $\boldsymbol{\alpha} = \mathbf{0}$ outside the attack interval. The tested nonzero bias magnitudes range from 0.0005 to 0.0050 pu, corresponding to 0.05%–0.5% current-feedback bias. Percentages refer to the nominal controller feedback base; no absolute ampere or torque calibration is inferred from them. The term severity denotes injected bias magnitude, not measured physical harm. Attack severity is measured by the largest absolute injected phase-current bias, $s = \|\boldsymbol{\alpha}\|_\infty$.

For binary FDIA detection, each window is labeled as $y_{\mathrm{fdia2}} = 1$ if $s > 0$ and $y_{\mathrm{fdia2}} = 0$ otherwise. For four-class

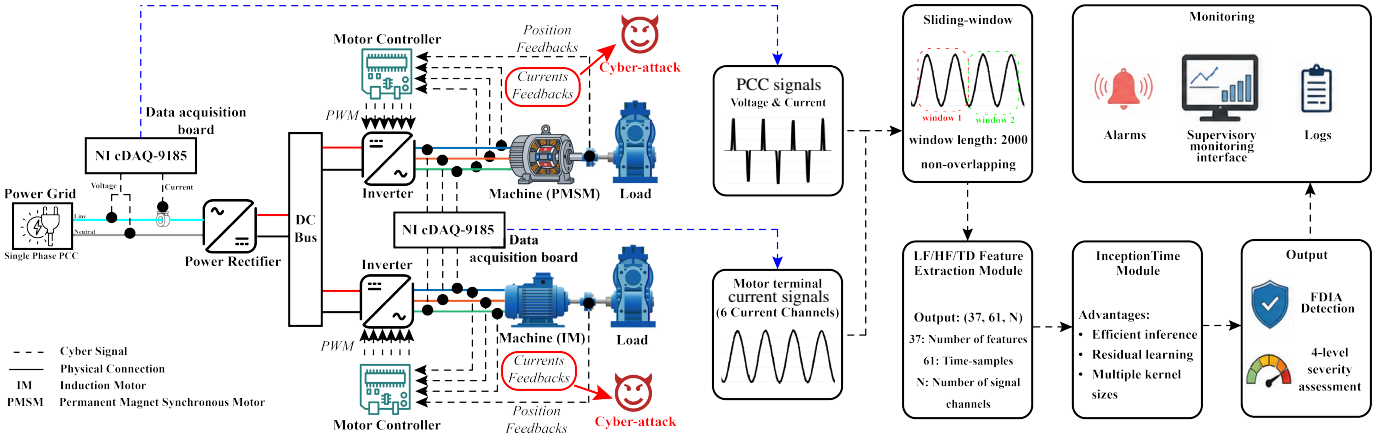


Fig. 1. Shared-rectifier, two-branch inverter-fed drive platform and monitoring locations. Motor-terminal currents and PCC voltage/current provide the monitoring inputs. The attack modifies controller-current feedback; the monitoring measurements are assumed trusted.

TABLE I
FDIA DATASET STRUCTURE.

Cases	Phases	Bias Pattern	Total
1–10	A	α_a: 0.0005–0.0050	180
11–20	A,B	$\alpha_a = \alpha_b$: 0.0005–0.0050	180
21–30	A,B	$\alpha_a = 0.0050$, α_b: 0.0005–0.0050	180
		Overall	540

Each case group is evaluated over 3 speeds, 3 load settings, and 2 motor types. For all cases, unspecified phase biases are zero.

TABLE II
MEASUREMENT STREAMS USED FOR MONITORING.

Stream	Measured signals	Channels
Motor-side	ACIM and PMSM phase currents	6
PCC-side	PCC voltage and current	2

severity assessment, the label is defined as

$$y_{\text{severity4}} = \begin{cases} 0, & s = 0 \quad \text{(normal)}, \\ 1, & 0.0005 \le s \le 0.0015 \quad \text{(low)}, \\ 2, & 0.0020 \le s \le 0.0035 \quad \text{(medium)}, \\ 3, & 0.0040 \le s \le 0.0050 \quad \text{(high)}. \end{cases} \tag{2}$$

The bins apply to the tested discrete bias values and are not standardized safety thresholds. Cases 21–30 have $s = 0.0050$ throughout because phase A remains fixed; all belong to the high-bias class. The resulting attack-window counts are 5,400 low, 7,200 medium, and 14,400 high. Table I summarizes the swept single-phase, equal two-phase, and fixed-plus-swept two-phase FDIA cases. Across three speeds, three load settings, and two motor types, the dataset contains 540 total attack cases. Each recording contains 30 consecutive 20-s episodes, with 10 s normal operation followed by 10 s FDIA operation. After the scheduled 600 s, the injected bias was zero and the motor continued running. All 6,054 complete post-schedule windows are retained as normal, giving 33,054 normal and 27,000 attack windows in total.

C. Data Acquisition and Operating Conditions

Electrical measurements were acquired from the testbed using a National Instruments cDAQ-9185 chassis and a custom data-collector interface board. The interface board routed synchronized motor-side and PCC-side analog channels to the acquisition system. Three brake-excitation settings were tested for each machine: 1–3 V for the ACIM branch and 2–4 V for the PMSM branch. Because torque was not directly measured, these brake-excitation voltages are used as load-condition labels. All channels were sampled at 10 kHz and stored in TDMS format. For learning-based analysis, each record was segmented into non-overlapping 0.2 s windows containing 2,000 samples. Non-overlap prevents reuse of raw samples between adjacent windows, but does not remove within-recording dependence. Therefore, each entire recording, including its 30 episodes and post-schedule normal segment, is assigned to one partition. For each motor, five recordings are used for training, two for validation, and two for testing. Three seeds (1234, 2234, and 3234) change both the recording assignment and initialization. All feature normalization statistics are fitted using training recordings only. For example, all windows from one ACIM speed/load recording stay together; an adjacent attack episode cannot appear in a different partition. The holdouts assess new recorded operating combinations on the same apparatus; recordings can recur across repeats. The resulting dataset dimensions, input configurations, severity-label distributions, and evaluation protocols are summarized in Table III.

IV. THE PROPOSED METHOD

A. LF/HF Feature Extraction Pipeline

To represent both slow electromechanical behavior and faster attack-induced disturbances, a two-scale LF/HF feature pipeline was applied independently to each measured channel. For every sample, a 2,000-point outer window was first extracted from the raw 10 kHz recording. Within this outer window, two parallel sliding analyses were performed. The LF branch uses a 500-sample window with a stride of 25 samples, while the HF branch uses a 250-sample window with

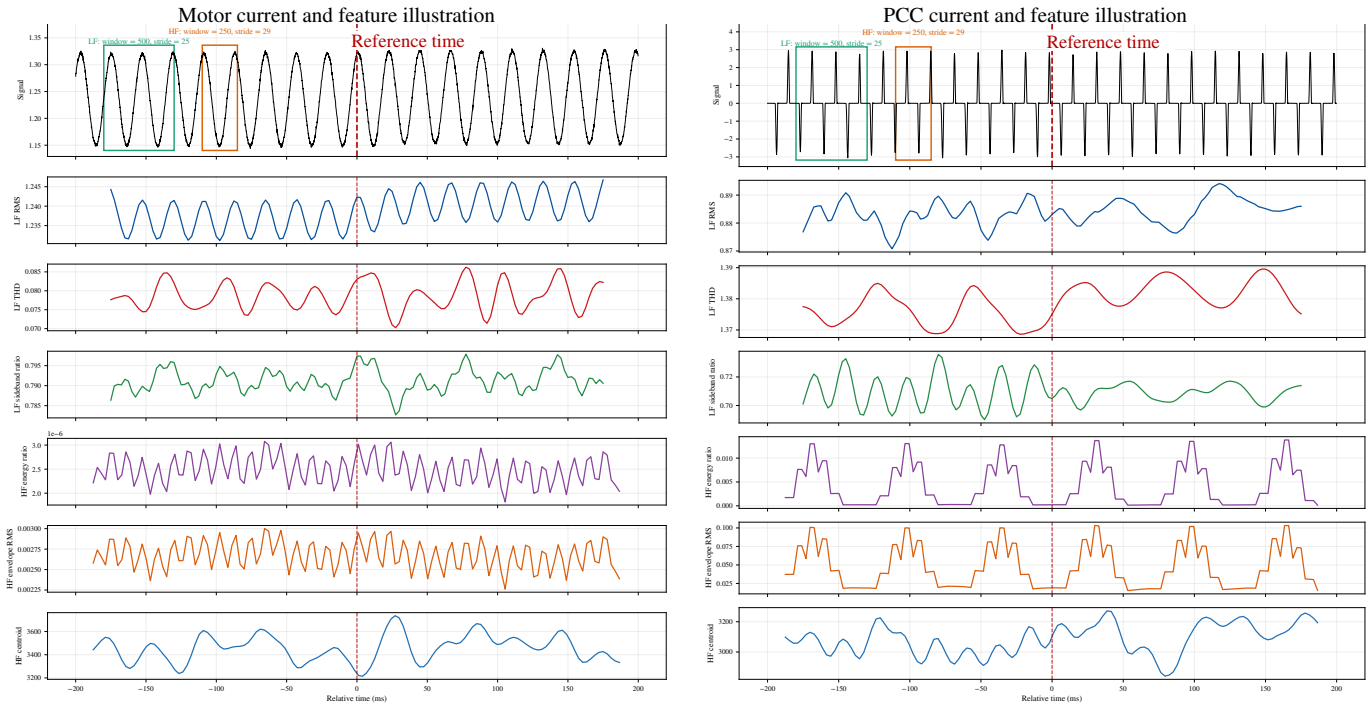


Fig. 2. Motor-current (left) and PCC-current (right) signal and feature illustrations. LF frames use 500 samples with stride 25; HF frames use 250 samples with stride 29. Zero denotes the display reference time. These descriptive traces illustrate feature extraction and do not establish attack-onset timing or detector performance. Signal amplitude is in recorded acquisition units.

a stride of 29 samples. The LF and HF settings both produce 61 time steps, enabling concatenation into one temporal feature tensor. Fig. 2 illustrates the motor-current and PCC-current waveforms together with representative LF/HF windows. Each trace is plotted relative to a display reference time. The overlaid frames show the two temporal scales used for feature extraction, while the lower panels show how selected features vary along the signal. These examples illustrate the extraction procedure; detection and severity performance are evaluated separately on held-out recordings.

In total, 37 features were extracted per channel, including 18 LF spectral features, 11 HF spectral/envelope features, and 8 TD features. The LF branch targets slow electromechanical changes, fundamental-frequency behavior, harmonic distortion, sideband effects, and low-band energy redistribution. The HF branch captures higher-frequency redistribution and envelope-related changes that may appear during small current-feedback FDIAs. The TD features provide complementary sensitivity to amplitude variation, impulsiveness, waveform asymmetry, and short-term signal complexity. Their relative contributions are evaluated through fixed-family ablations; improved separability is assessed empirically rather than assumed for every feature family.

B. Inception-Based Detection Model

The extracted LF/HF tensors were modeled using a one-dimensional InceptionTime architecture [15] for multivariate time-series classification. The input dimension is $(61, 222)$ for the motor stream and $(61, 74)$ for the PCC stream. As illustrated in Fig. 3, the network uses bottleneck compression, parallel one-dimensional convolutional kernels of lengths 39, 19, and 9, residual shortcuts, global average pooling, and a dense output layer. Motor and PCC models used stream-specific hyperparameters selected by validation balanced accuracy. The Inception-based design is well suited to the proposed LF/HF representation because the extracted features retain temporal ordering across the 61-step sequence. This preserves temporal interactions among LF, HF, and TD features.

C. Architecture of Online FDIA Detection and Attack-Severity Estimation

The proposed pipeline supports online monitoring from trusted motor-terminal currents or PCC voltage/current measurements. Figure 4 illustrates the overall architecture of the proposed monitoring pipeline. In the intended monitoring setting, fixed-length windows of electrical measurements are acquired from either the motor terminals or the PCC, passed through the LF/HF feature-extraction stage, and then evaluated by the selected learning model to produce anomaly or severity decisions. The measurement, feature-extraction, and inference stages are modular, allowing the same pipeline to operate with motor-side or PCC inputs and supporting feature pruning, reduced-input models, and cross-motor adaptation. The framework provides alarms, severity indicators, and logs for supervisory systems; automatic mitigation is outside the scope. The same architecture is used for binary detection and severity classification. Binary detection is the primary PCC deployment task; severity estimates provide secondary diagnostic information. The experimental results in later sections show that motor-terminal sensing provides cleaner discriminative

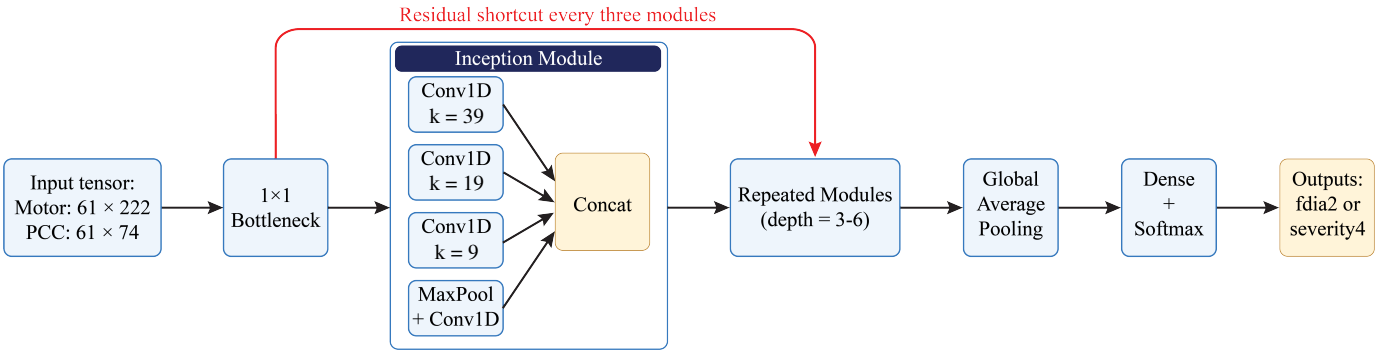


Fig. 3. InceptionTime architecture for `fdia2` and `severity4`, with convolution lengths 39, 19, and 9. Each module has three temporal convolutions and a max-pooling projection branch, followed by concatenation, batch normalization (BN), and ReLU. A projected residual shortcut is added after each group of three modules; depth and other settings are selected separately for each task, stream, and repeat.

TABLE III
DATASET DIMENSIONS AND RECORDING-DISJOINT EVALUATION PROTOCOLS.

Parameter	Value / configuration
Data dimensions	
Complete windows	60,054; 2,000 samples (0.2 s) each
Recordings / episodes	18 recordings; 540 scheduled episodes
Motor distribution	ACIM: 30,204; PMSM: 29,850
Normal / attack windows	33,054 normal; 27,000 attack
Normal tail windows	6,054; included with parent recording
Input configuration	
Features per channel	18 LF; 11 HF; 8 TD
Temporal feature steps	61 per outer window
Motor input	61 × 222; six current channels
PCC input	61 × 74; voltage and current
Severity labels (`severity4`)	
Normal / low	33,054 / 5,400 windows
Medium / high	7,200 / 14,400 windows
Protocol 1: mixed-motor holdouts	
Recordings per motor	5 training; 2 validation; 2 test
Repeat 1 (seed 1234)	33,718 training; 13,279 validation; 13,057 test windows
Repeat 2 (seed 2234)	33,275 training; 13,250 validation; 13,529 test windows
Repeat 3 (seed 3234)	33,560 training; 13,227 validation; 13,267 test windows
Protocol 2: cross-motor adaptation	
Source recordings	6 training; 3 validation
Target recordings	6 adaptation-pool; 3 held-out test
Target adaptation budgets	1/5/10/20%: 2/9/18/36 complete episode groups, respectively, from 180 groups

Each recording and all its episodes remain in one partition within a repeat.

TABLE IV
TOP FIVE CONFIGURATIONS BY VALIDATION BALANCED ACCURACY ON `severity4`, SEED 1234.

Rank	D	F	BN	Dr	LR	WD	BS	Val. BA
MOTOR stream								
1	3	16	8	0.50	0.001	10^{-4}	64	**93.26**
2	3	24	8	0.20	0.0005	5×10^{-5}	64	92.51
3	6	24	16	0.35	0.001	5×10^{-5}	128	92.48
4	3	32	8	0.20	0.001	5×10^{-5}	128	92.37
5	4	16	8	0.50	0.001	5×10^{-5}	128	92.21
PCC stream								
1	6	32	32	0.50	0.001	10^{-4}	64	**52.25**
2	6	48	32	0.35	0.001	5×10^{-5}	64	51.94
3	3	16	32	0.50	0.001	5×10^{-5}	64	51.44
4	4	48	32	0.50	0.001	10^{-4}	128	51.26
5	6	24	32	0.35	0.0005	10^{-4}	128	51.14

D: depth; F: filters; BN: bottleneck; Dr: dropout; LR: learning rate; WD: weight decay; BS: batch size. Bold: highest validation BA within each stream. Test scores are not used for ranking.

signatures, whereas PCC sensing offers a less intrusive sensing option for the tested setup.

D. Training Details and Hyperparameters

A fixed sample of 100 candidate configurations was evaluated independently for each task (binary detection or four-class severity), sensing stream, and recording split, giving 1,200 main search fits. Candidate ranges were depth 3/4/6, filters 16/24/32/48, bottleneck width 8/16/32, dropout 0.20/0.35/0.50, learning rate 5×10^{-4} or 10^{-3}, weight decay 5×10^{-5} or 10^{-4}, and batch size 64 or 128. The sampled candidate set, rather than every Cartesian combination, was searched. Training used Adam, inverse-frequency weighted cross-entropy, a 60-epoch cap, and early stopping after 10 epochs without validation-balanced-accuracy improvement. A plateau scheduler halved the learning rate after three unimproved epochs, down to 10^{-6}. Checkpoints and hyperparameters were selected only by validation balanced accuracy; ties followed the fixed candidate order. Table IV illustrates the ranking for one specified split.

Reported test classification metrics are mean ± sample standard deviation over the three holdouts. Precision, recall, and F1 are macro-averaged; balanced accuracy is the mean class recall. Because the same recordings can recur across repeats, these standard deviations describe split/initialization sensitivity rather than uncertainty from independent physical replications. Test results were not used to select configurations or feature subsets. One-factor sensitivity experiments changed depth to 3/9, dropout to 0.1/0.4, or learning rate to $3\times10^{-4}/3\times10^{-3}$ around each selected configuration; all other settings were retained (Fig. 10).

E. Domain Adaptation Strategy: Source-Only vs Few-Shot Fine-Tuning

Source-only transfer was evaluated in both directions, ACIM → PMSM and PMSM → ACIM. For each direction, six source

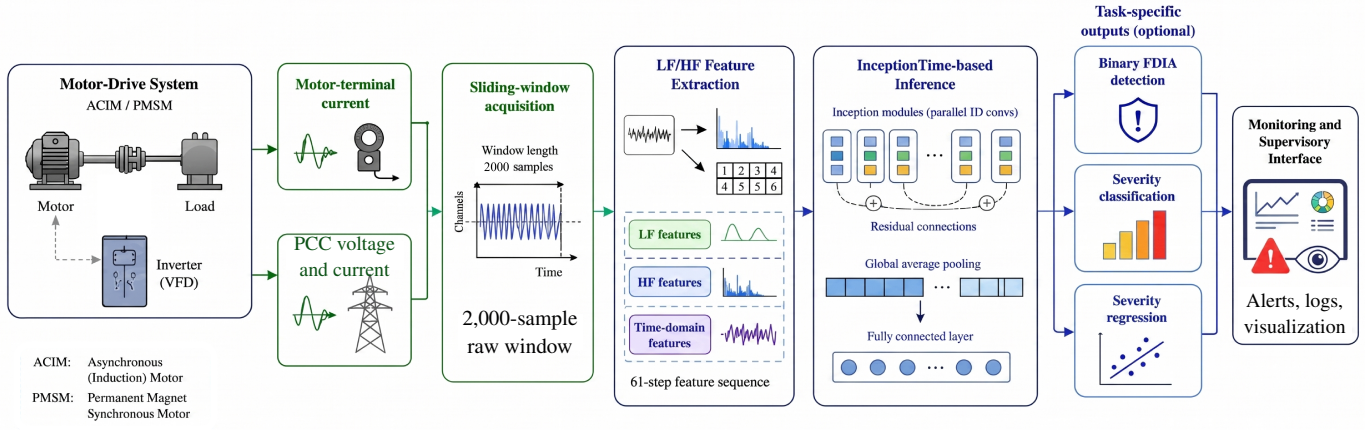


Fig. 4. Monitoring pipeline for motor-side and PCC-side sensing. Motor-terminal currents or PCC voltage/current enter a 2,000-sample (0.2-s) raw window, followed by the 61-step feature representation and task-specific inference. Regression is an auxiliary injected-bias estimate. The interface provides alerts, logs, and visualization; automatic mitigation is not evaluated.

recordings were used for training and three for validation. The same 100-candidate search was performed using source validation only, independently of mixed-domain model selection. Six target recordings formed the adaptation pool and the remaining three formed the target test set. Thus, no source or target recording crosses its fitting/evaluation boundary.

Nominal adaptation budgets of 1%, 5%, 10%, and 20% select 2, 9, 18, or 36 complete episode groups from the 180-group target pool; the first realized group fraction is 1.11%. For sampling only, each post-schedule normal tail is attached to its parent recording's final episode group while retaining its normal label and actual timestamp. Consequently, window fractions can differ from nominal group fractions. Fine-tuning updates the source model for 20 fixed epochs at 0.2 times its selected learning rate, with source normalization frozen. No target validation labels or target-test labels are used. All conditions use three seeds.

TABLE V
MIXED ACIM/PMSM PERFORMANCE ON HELD-OUT RECORDINGS (%).

Task	Acc.	BA	Prec.	Recall	F1
PCC stream					
`fdia2`	80.78 (±2.95)	80.29 (±3.03)	81.04 (±3.14)	80.29 (±3.03)	80.42 (±2.96)
`severity4`	61.00 (±1.32)	55.62 (±2.94)	52.93 (±2.36)	55.62 (±2.94)	52.96 (±2.15)
Motor type	94.40 (±9.70)	94.35 (±9.79)	95.35 (±8.06)	94.35 (±9.79)	94.27 (±9.92)
MOTOR stream					
`fdia2`	**87.41** (±11.13)	**88.26** (±9.99)	**89.90** (±7.84)	**88.26** (±9.99)	**87.23** (±11.39)
`severity4`	**87.64** (±8.87)	**90.81** (±4.29)	**86.09** (±9.44)	**90.81** (±4.29)	**86.06** (±8.95)
Motor type	**99.94** (±0.11)	**99.94** (±0.11)	**99.94** (±0.10)	**99.94** (±0.11)	**99.94** (±0.11)

Cells show mean above (± sample SD), over three holdouts; this is not a range or confidence interval. BA: balanced accuracy. Precision, recall, and F1 are macro-averaged. Bold compares streams for the same task and metric, not different tasks.

V. EXPERIMENTAL RESULTS

A. In-Domain Baselines: Motor vs PCC

Table V reports mixed-domain performance with both ACIM and PMSM represented in every partition but with entire recordings held out. Motor-side mean balanced accuracy exceeds PCC performance for detection (88.26% versus 80.29%) and severity (90.81% versus 55.62%). Motor detection is sensitive to the held-out operating combinations: accuracies across the three splits are 88.83%, 97.77%, and 75.64%. This variability limits a claim of uniformly reliable performance across operating conditions.

For the separate binary detectors, normal false-alarm rates are $21.15 \pm 21.77\%$ for motor sensing and $14.82 \pm 5.06\%$ for PCC sensing; missed-attack rates are $24.61 \pm 7.17\%$ for PCC and $2.32 \pm 1.79\%$ for motor sensing. Therefore, mean accuracy alone does not establish a deployment-ready alarm rate.

For the four-class PCC model, exact low/medium/high recall is 38.33%/50.58%/70.79%, whereas detecting those windows as any attack yields 61.39%/85.08%/88.09%. The complementary severity-conditioned attack recall is defined as

$$R_{\text{attack}|k} = \frac{\sum_{j=1}^{3} C_{kj}}{\sum_{j=0}^{3} C_{kj}}, \qquad k \in \{1,2,3\}, \tag{3}$$

where C is the confusion matrix. Exact-severity recall is $C_{kk} / \sum_j C_{kj}$. These measures answer different questions: an attack can be detected while its magnitude bin is incorrect. The same PCC severity model has a 37.23% mean normal false-alarm rate and 71.40% mean accuracy after collapsing its predictions to normal/attack. It is distinct from the separately trained binary detector. Fig. 6 and Table XIII report these complementary quantities without replacing four-class performance.

B. Comparison with Deep and Classical ML Baselines

ResNet, LSTM-FCN, Random Forest, XGBoost, and MultiRocket [16] were evaluated on the same splits and corrected targets. The LSTM-FCN baseline denotes the implementation actually used, without claiming an additional squeeze-and-excitation MLSTM-FCN architecture. ResNet used 40 epochs

TABLE VI
PCC BASELINE COMPARISON ON FDIA2 AND SEVERITY4 (%).

Method	Accuracy	BA	Macro F1
fdia2			
InceptionTime	**80.78** (±2.95)	**80.29** (±3.03)	**80.42** (±2.96)
ResNet	73.91 (±5.78)	73.99 (±5.33)	73.76 (±5.66)
LSTM-FCN	72.57 (±4.89)	72.52 (±4.31)	72.35 (±4.66)
Rand. Forest	70.43 (±1.00)	68.69 (±0.74)	68.45 (±0.85)
XGBoost	75.37 (±0.81)	73.84 (±1.11)	73.97 (±1.17)
MultiRocket	70.10 (±2.13)	69.55 (±2.82)	69.35 (±2.39)
severity4			
InceptionTime	61.00 (±1.32)	**55.62** (±2.94)	52.96 (±2.15)
ResNet	65.18 (±5.35)	50.41 (±5.44)	51.15 (±5.71)
LSTM-FCN	66.60 (±1.72)	52.38 (±2.12)	**53.23** (±1.94)
Rand. Forest	67.36 (±0.17)	43.49 (±2.18)	45.07 (±2.59)
XGBoost	**70.45** (±0.74)	47.17 (±2.02)	49.67 (±2.17)
MultiRocket	58.41 (±3.34)	44.92 (±3.46)	44.28 (±2.52)

Mean above (± SD). Bold: highest mean for each metric within a task; it does not indicate statistical significance.

and batch size 64; LSTM-FCN used 40 epochs, batch size 128, eight LSTM units, dropout 0.5, and convolutional widths 64/128/64. Trees used the flattened $61 \times F$ tensors: Random Forest compared 300/600 trees and unrestricted/20 depth with balanced-subsample weights; XGBoost compared depth 4 with learning rate 0.05 and depth 6 with learning rate 0.03, using 1,500 estimators and 0.8 row/column subsampling. MultiRocket used approximately 50,000 transformed features and ridge penalties 0.1/1/10/100. Available candidate settings were selected using validation balanced accuracy. Search budgets differ between model families, so this is a comparison under the stated settings rather than an equal-budget optimization study.

As shown in Table VI, InceptionTime has the highest mean PCC balanced accuracy for both tasks, and the highest mean accuracy and macro F1 for binary detection. For severity, XGBoost has higher raw accuracy (70.45% versus 61.00%), while LSTM-FCN has slightly higher macro F1 (53.23% versus 52.96%). InceptionTime's severity advantage is confined to mean balanced accuracy (55.62% versus at most 52.38% among the evaluated alternatives). Unequal class counts make these distinctions material; no statistical-significance claim is made from three overlapping repeated holdouts.

C. Cross-Domain Transfer Without Adaptation

Source-only binary transfer is unreliable on held-out target recordings (Table VII). Balanced accuracy is 50.60% and 51.58% for ACIM-to-PMSM motor and PCC sensing, respectively. In the reverse direction it is 36.08% for motor and 56.00% for PCC sensing. The below-chance motor result demonstrates systematic domain mismatch rather than useful transferable discrimination. Accuracy must be interpreted jointly with balanced accuracy because predicting the more common class can exceed 50% raw accuracy without effective attack detection. Target-domain fine-tuning tests whether labeled target episodes alleviate this mismatch.

TABLE VII
SOURCE-ONLY FDIA2 TRANSFER WITHOUT TARGET ADAPTATION (%).

Direction	Accuracy	BA	Macro F1
MOTOR stream			
ACIM → PMSM	55.21 (±1.14)	50.60 (±0.73)	37.71 (±2.37)
PMSM → ACIM	36.22 (±4.23)	36.08 (±4.11)	33.89 (±4.62)
PCC stream			
ACIM → PMSM	55.88 (±0.64)	51.58 (±1.52)	41.13 (±5.84)
PMSM → ACIM	51.94 (±6.54)	56.00 (±5.69)	43.62 (±11.38)

Mean above (± SD). Directions are different target-domain tests, so no best direction is highlighted.

TABLE VIII
FEW-SHOT FDIA2 ADAPTATION BY LABELED TARGET-DOMAIN FINE-TUNING: BALANCED ACCURACY (%).

Direction	Source only	1%	5%	10%	20%
MOTOR stream					
ACIM → PMSM	50.60 (±0.73)	50.43 (±1.74)	53.83 (±1.14)	57.51 (±7.24)	**75.99** (±1.73)
PMSM → ACIM	36.08 (±4.11)	64.35 (±9.23)	73.93 (±8.78)	88.04 (±5.36)	**92.63** (±3.72)
PCC stream					
ACIM → PMSM	51.58 (±1.52)	54.01 (±0.12)	54.62 (±1.76)	54.04 (±1.35)	**56.30** (±3.23)
PMSM → ACIM	56.00 (±5.69)	77.43 (±10.43)	84.22 (±4.62)	86.72 (±4.37)	**91.38** (±2.06)

Mean above (± SD). Bold: highest observed mean within each direction and stream; no test-based configuration selection. Budgets refer to the target episode-group pool.

D. Domain Adaptation Results: 1/5/10/20%

Table VIII reports fdia2 balanced accuracy before and after target-domain fine-tuning. At 20%, motor fine-tuning reaches 75.99% for ACIM-to-PMSM and 92.63% for PMSM-to-ACIM transfer. PCC fine-tuning reaches 56.30% and 91.38%, respectively, preserving a pronounced direction asymmetry.

The benefit is direction dependent. At the 1% budget, motor ACIM-to-PMSM performance remains near the source-only result, whereas PMSM-to-ACIM performance improves substantially. Increasing the target budget generally helps, but the PCC ACIM-to-PMSM sequence is not monotonic. These comparisons measure improvement over the source-only model; they do not establish superiority over training directly on the same target labels.

E. Error Analysis: Confusion Matrices and FP/FN Behavior

Fig. 5 shows the full error distributions. Binary motor sensing combines high attack recall with a substantial and variable normal false-alarm rate; PCC sensing misses more attacks. For the PCC severity model, mistakes include both normal/attack confusion and confusion among nonzero bias bins. Therefore,

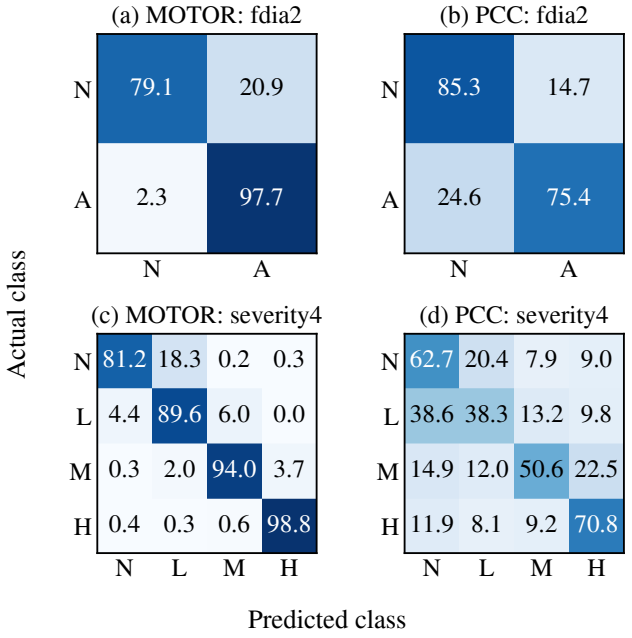


Fig. 5. Confusion matrices for `fdia2` and `severity4`. Rows are actual and columns predicted classes. Percentages normalize pooled prediction counts within each row across three holdouts; repeated appearances of a recording are not independent replications. N/L/M/H denote normal/low/medium/high bias.

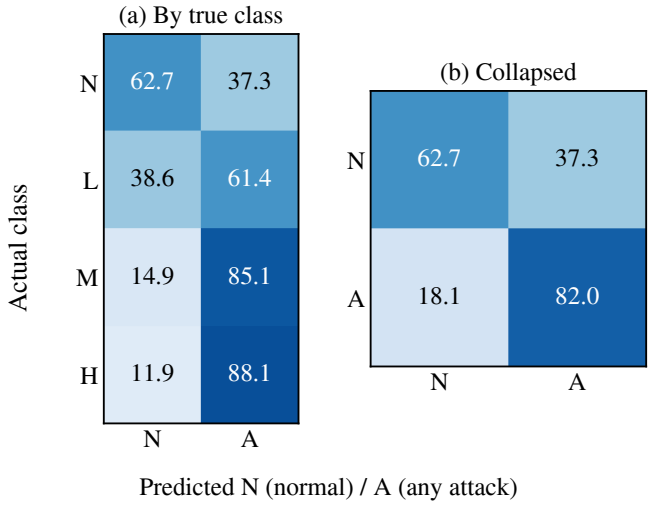


Fig. 6. PCC `severity4` outputs interpreted as (a) normal/any-attack decisions by actual class and (b) a collapsed binary decision. Row percentages pool three holdouts. These decisions come from the severity model, not the separately trained `fdia2` detector.

the errors should not be characterized solely as confusion between adjacent levels. Fig. 6 presents the complementary PCC decisions here. A low-bias attack predicted as medium or high is detected but its exact severity is incorrect. Normal false alarms must be considered alongside any-attack recall. Mean and SD values are listed in Appendix A-C.

Fig. 7 shows a 200-s overview and a 22-s transition detail from a held-out ACIM recording. The recording was selected by a fixed alphabetical rule, not for visual agreement with the schedule. The trace exposes both successful detections and false alarms. The complete recording, including its normal tail, and a corresponding held-out PMSM trace are supplied with the revision materials.

VI. Latency and Reduced-Feature Inference Analysis

Practical PCC monitoring requires low latency and stable online throughput. This section reports per-window latency, throughput, pruning, and reduced-feature effects. In particular, the analysis focuses on end-to-end processing latency, online throughput, feature-pruning effects, and practical considerations for deployment at the PCC.

A. Inference Latency and Throughput on CPU/GPU

Runtime was measured on the Intel Core Ultra 9 285K CPU and NVIDIA RTX 5090 GPU using the seed-1234 binary models. Each variant used 10 warm-up executions and 50 timed repetitions. The benchmark separates feature extraction, normalization/transfer/inference, and full processing, reporting median and 95th-percentile latency (Table IX). End-to-end processing excludes the 200-ms acquisition window; it is not an attack-onset-to-alarm delay. CPU and GPU-assisted versions use the same selected models.

TABLE IX
GPU-ASSISTED PER-WINDOW LATENCY AND PROCESSING RATE.

Statistic	Feature extract. (ms)	Norm./transfer + inference (ms)	Total processing (ms)	Rate (windows/s)
PCC stream				
Median	21.64	0.72	22.37	44.69
95th pct.	23.11	0.85	23.93	–
MOTOR stream				
Median	64.99	1.33	66.33	15.08
95th pct.	66.19	1.42	67.55	–

Rate is the reciprocal of median total processing time. Stages are timed within each iteration; marginal percentiles need not sum. Acquisition (200 ms) is excluded. CPU measurements are supplied with the results.

Full-feature GPU-assisted processing takes 22.37 ms for PCC and 66.33 ms for motor sensing, with 95th percentiles of 23.93 and 67.55 ms. Both measured processing times are below the 200-ms window interval. Reciprocal median processing rates are approximately 44.69 and 15.08 windows/s, respectively; these are workstation processing capacities, not measured embedded worst-case guarantees. Acquisition, scheduling, and any alarm-persistence logic must be included in a deployed timing budget.

B. Feature Pruning for Low-Latency Anomaly Detection

Binary-model features were ranked by validation-set permutation importance: the decrease in balanced accuracy when one feature trajectory is permuted across validation windows, repeated three times. Test labels were not used for ranking. Within each split, global top-75%, top-50%, and top-25% subsets and per-channel top-15/top-10 subsets were retrained with the full model's selected architecture and training protocol. Fig. 8 summarizes validation importance across repeats;

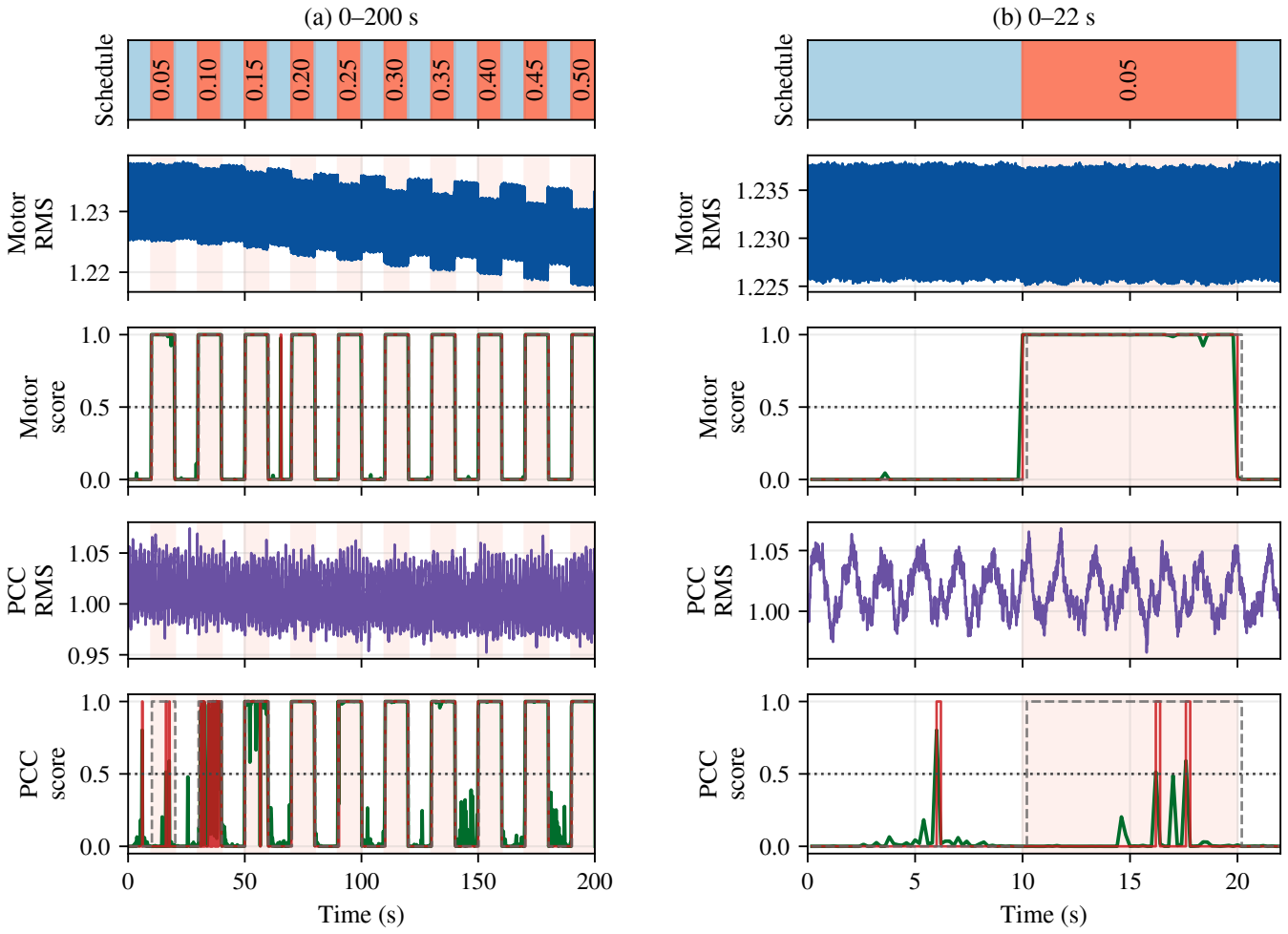


Fig. 7. Held-out ACIM detection: schedule, motor RMS, motor decision, PCC RMS, and PCC decision. (a) First 200 s; (b) first 22 s, including the first low-bias attack. The recording is the first alphabetically among seed-1234 test recordings. Curves use saved held-out predictions at window ends; RMS uses 500-sample frames at stride 25. Green: attack probability; red: predicted label; dashed gray: true label; dotted line: 0.5 threshold. Shading denotes scheduled attacks. Complete-recording and PMSM plots accompany the paper.

correlated features can share predictive information, so these scores do not identify causal attack mechanisms.

Table X shows that fewer retained features do not guarantee faster extraction. The PCC top-25% subset retains 18 of 74 features and approximately preserves balanced accuracy (80.29%), but processing remains 22.42 ms versus 22.37 ms for full features. The motor top-25% subset retains 56 of 222 features, gives 86.83% balanced accuracy versus 88.26% for full input, and reduces processing from 66.33 to 56.41 ms (1.18×). The extractor skips an LF, HF, or TD family only when that entire family is absent from a channel; otherwise it computes every statistic in that family before selecting the requested columns. In the seed-1234 PCC top-25% subset, all three families remain active on both channels: each family routine is called 122 times per window, exactly as for full input. Motor top-25% removes the HF family on two channels, reducing HF calls from 366 to 244. Thus, the current implementation provides family-level computational pruning, not individual-feature computation. The nearly unchanged PCC latency reflects this implementation limit; it does not establish that a feature-level optimized extractor could not be faster. All subsets are reported; no test-best subset replaces the primary full-feature model.

C. LF/HF/TD Ablation for Binary Anomaly Detection

All seven LF/HF/TD combinations were evaluated by retraining with the selected full-feature architecture fixed within each task, stream, and split. Table XI reports binary results. For PCC, LF+Time reaches 80.03% balanced accuracy, close to the full input at 80.29%, while reducing GPU-assisted processing from 22.37 to 12.32 ms (1.82×). Thus, the measured marginal PCC benefit of HF features does not establish that their extra computation is necessary for binary detection on this dataset.

For motor sensing, time-only reaches 92.89% balanced accuracy and HF+Time reaches 94.11%, compared with 88.26% for the full input. Time-only processing takes 16.28 ms (4.07× faster than full). HF-only performs poorly for both streams (63.76% motor and 64.49% PCC balanced accuracy), but its combination with time features is useful for motor sensing. These are observed ablation results, not test-based selection

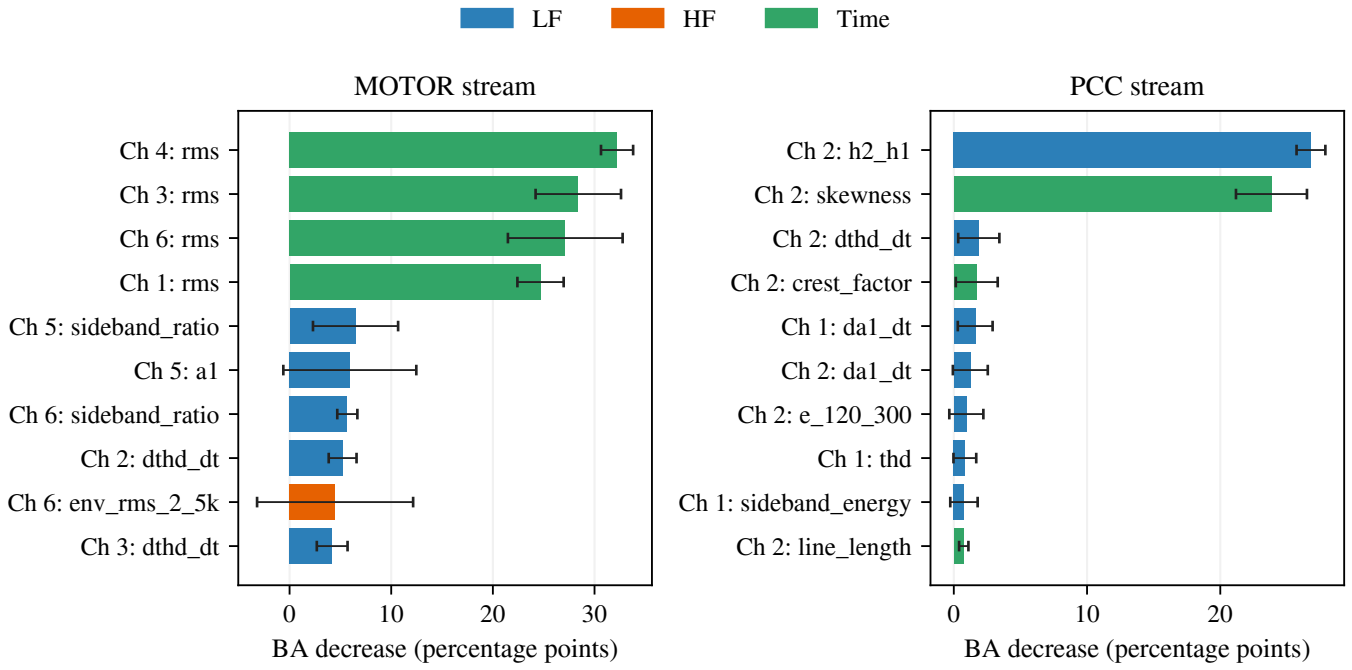


Fig. 8. Top ten channel-specific features ranked by mean validation permutation importance. Blue, orange, and green bars denote LF, HF, and time-domain features, respectively. Bar length is the mean balanced-accuracy decrease; black whiskers indicate one SD across three holdouts. Channels 1–3/4–6 are ACIM/PMSM currents; PCC channels 1/2 are voltage/current.

TABLE X
FEATURE PRUNING FOR FDIA2: PERFORMANCE AND RUNTIME.

Subset	No.	BA (%)	F1 (%)	Extract. (ms)	Total (ms)	Speedup (×)
PCC stream						
Full	74	80.29 (±3.03)	80.42 (±2.96)	21.64	**22.37**	1.00
75%	56	78.28 (±3.07)	78.41 (±3.37)	21.77	22.46	1.00
50%	37	78.87 (±2.79)	79.10 (±2.97)	21.83	22.58	0.99
25%	18	80.29 (±2.14)	80.58 (±2.29)	21.70	22.42	1.00
15/ch.	30	77.59 (±4.75)	77.74 (±4.94)	21.80	22.49	0.99
10/ch.	20	**80.47** (±2.79)	**80.69** (±2.80)	21.94	22.68	0.99
MOTOR stream						
Full	222	88.26 (±9.99)	87.23 (±11.39)	64.99	66.33	1.00
75%	166	87.95 (±11.38)	87.19 (±12.42)	65.65	67.02	0.99
50%	111	89.28 (±11.33)	88.20 (±12.87)	65.14	66.49	1.00
25%	56	86.83 (±10.93)	85.44 (±12.39)	55.11	**56.41**	1.18
15/ch.	90	**92.10** (±9.71)	**91.44** (±10.88)	60.80	62.13	1.07
10/ch.	60	88.18 (±9.28)	87.20 (±10.62)	55.45	56.77	1.17

Performance: mean above (± SD). Timing: seed-1234 GPU-assisted median; 95th percentiles are supplied in the results. Bold: highest BA/F1 or lowest total latency within a stream. Only whole unused feature families are skipped.

of a new primary model. Severity ablations and complete per-repeat measurements are supplied with the revision materials.

VII. Discussion

A. Why Adaptation Helps More in One Direction

The PCC results retain a strong direction asymmetry: at 20% target labels, PMSM-to-ACIM fine-tuning reaches 91.38% balanced accuracy, while ACIM-to-PMSM reaches 56.30%. Source-only predictions also show different class biases: for PCC ACIM-to-PMSM, the mean attack miss rate is 92.71%; in the reverse direction the normal false-alarm rate is 86.03%. These biases show that source-domain decision boundaries and operating distributions do not transfer reliably. They do not isolate a causal mechanism for the asymmetry.

The adaptation results therefore support direction-specific target calibration. In particular, the strong PCC PMSM-to-ACIM gain is not reproduced in the reverse direction. Source-only and fine-tuned results quantify the benefit of updating a source model with target labels; they do not isolate the contribution of source pretraining from the benefit of those labels.

B. PCC vs Motor Tradeoffs

Motor sensing provides higher mean class-balanced performance on this dataset, particularly for distinguishing injected-bias bins. PCC sensing reduces terminal instrumentation but aggregates the responses of both drive branches and their shared converter interface. For the four-class PCC model, relatively high any-attack recall at medium/high bias coexists with low exact-severity recall and frequent false alarms. The separate binary detector is the stronger primary PCC alarm model in these experiments.

TABLE XI
LF/HF/TD ABLATION FOR FDIA2 (%).

Features	No.	Accuracy	BA	Macro F1
PCC stream				
		80.78	**80.29**	**80.42**
Full	74	(±2.95)	(±3.03)	(±2.96)
		74.03	73.56	73.59
LF only	36	(±3.84)	(±3.13)	(±3.33)
		64.50	64.49	64.21
HF only	22	(±4.62)	(±4.56)	(±4.71)
		72.18	70.66	70.65
Time only	16	(±1.73)	(±1.77)	(±1.76)
		68.66	68.15	68.18
LF + HF	58	(±4.60)	(±4.26)	(±4.39)
		80.49	80.03	80.16
LF + Time	52	(±1.73)	(±1.79)	(±1.73)
		69.49	67.85	67.72
HF + Time	38	(±3.66)	(±3.11)	(±3.06)
MOTOR stream				
		87.41	88.26	87.23
Full	222	(±11.13)	(±9.99)	(±11.39)
		81.61	80.54	80.94
LF only	108	(±3.08)	(±2.55)	(±2.77)
		63.32	63.76	63.14
HF only	66	(±4.96)	(±4.28)	(±4.81)
		92.61	92.89	92.58
Time only	48	(±5.51)	(±4.96)	(±5.50)
		80.33	79.91	80.05
LF + HF	174	(±7.88)	(±7.06)	(±7.60)
		90.70	91.17	90.65
LF + Time	156	(±8.32)	(±7.45)	(±8.35)
		93.87	**94.11**	**93.85**
HF + Time	114	(±4.56)	(±4.04)	(±4.54)

Mean above (± SD). Bold: highest mean within each stream; these test comparisons do not select a replacement primary model.

The motor-type task is an auxiliary diagnostic of distinguishable recording domains, not proof of a causal motor-physics explanation: the motor families also have different load-setting ranges and share only one installation per family. Attack-only regression likewise estimates injected bias rather than torque or damage (Appendix A-E). For efficiency, the PCC LF+Time representation offers a measured alternative to full features with similar balanced accuracy. Neither classification scores nor processing latency alone establish safe automatic mitigation.

C. Limitations

The evidence is limited to additive, piecewise-constant current-feedback bias on one two-branch laboratory platform. Monitoring sensors are assumed uncompromised; coordinated attacks on those sensors, replay, denial of service, command attacks, and stealthy time-varying perturbations were not tested. Three fixed speed settings and three load settings were recorded for each machine, but dedicated speed ramps, load steps, and other dynamic operating conditions were not evaluated. Brake excitation identifies operating settings; calibrated shaft torque and mechanical harm were not measured.

PCC signatures can depend on the shared rectifier, downstream inverter topology, control behavior, modulation, sensor transfer functions, and supply impedance. These factors were not varied independently in the dataset. Consequently, no robustness claim is made for other converter topologies, modulation strategies, grids, or installations. Whole-recording holdouts remove within-repeat recording/episode overlap, but remain a retrospective evaluation on reused data from the same apparatus. Three repeated holdouts do not provide independent hardware replication or a strong significance test.

The adaptation study requires target labels and uses fixed training budgets; unsupervised adaptation is outside its scope. PCC severity remains weak, and normal false alarms are material even when attack recall is high. Finally, measured workstation processing times exclude acquisition and do not establish embedded worst-case latency or a tested closed-loop mitigation system.

VIII. CONCLUSION

This paper assessed nonintrusive FDIA monitoring in a heterogeneous ACIM/PMSM inverter-fed drive platform using recording- and episode-disjoint evaluation. Across three holdouts, binary accuracy was 87.41% for motor-side sensing and 80.78% for PCC sensing. Four-class injected-bias balanced accuracy was 90.81% and 55.62%, respectively, demonstrating a substantial limitation of PCC severity estimation. Severity-conditioned attack recall complements exact-bin recall but must be interpreted alongside false alarms. Cross-motor source-only transfer was unreliable, and supervised adaptation benefits depended on direction and sensing location; the source-only comparison does not establish a universal benefit of pretraining. PCC LF+Time retained approximately full-feature binary balanced accuracy while reducing processing from 22.37 to 12.32 ms on the tested workstation. The results support further development of PCC monitoring for additive feedback-bias attacks while delimiting its accuracy, calibration, and deployment constraints.

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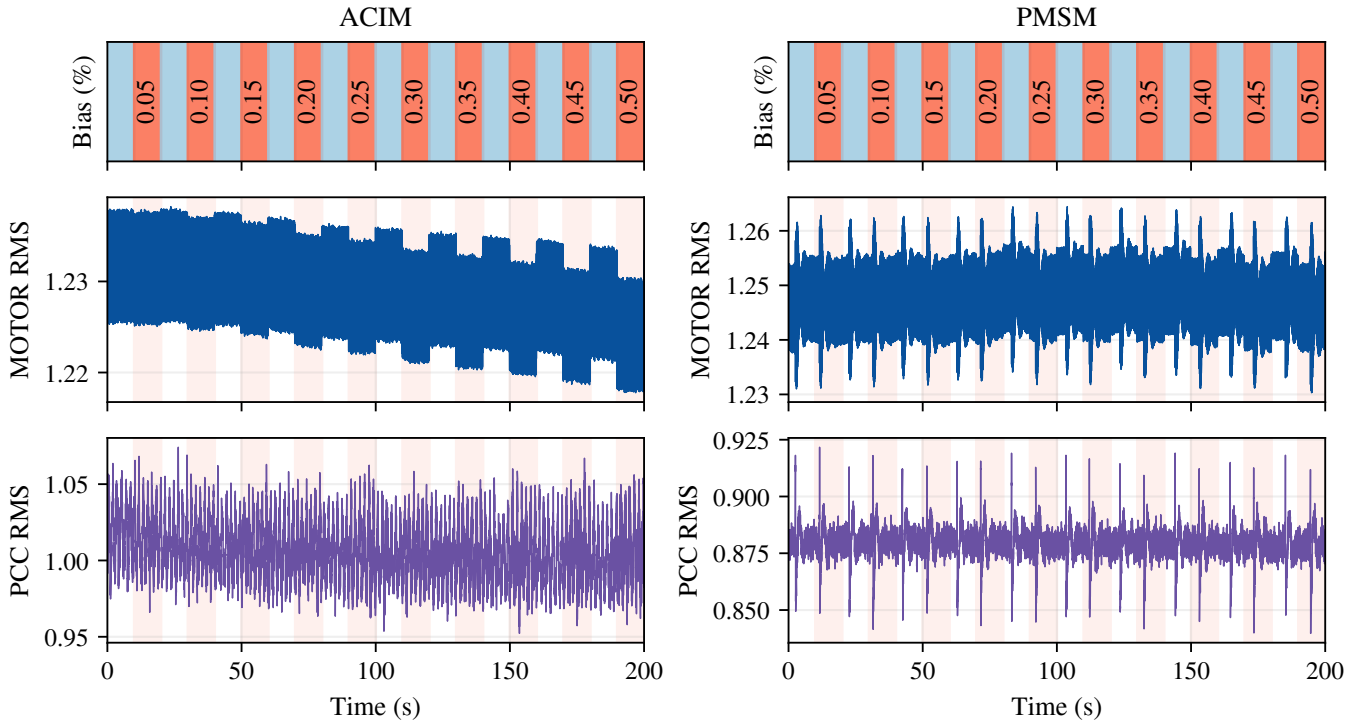


Fig. 9. First 200 s of prespecified held-out ACIM and PMSM recordings, showing the schedule and current RMS. Blue and red schedule blocks denote normal and attack intervals; numbers give maximum injected bias (%). Middle/bottom: motor/PCC RMS over 500-sample frames at stride 25, in acquisition units. These are descriptive electrical traces, not calibrated torque measurements.

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Appendix A
Additional Experimental and Diagnostic Results

This appendix provides representative FDIA recordings, the extracted feature definitions, severity-conditioned detection metrics, hyperparameter sensitivity, and auxiliary bias regression. Complete numeric results and per-repeat configurations accompany the manuscript.

A. Representative FDIA Episodes

Figure 9 shows representative consecutive FDIA episodes applied to the ACIM and PMSM drive systems. In both examples, the top panel shows the scheduled FDIA severity level, while the middle and bottom panels show the corresponding motor-current and PCC-current RMS responses during the first 200 s of each selected recording. RMS uses 500-sample frames at stride 25, matching the time-domain RMS definition. Complete-recording diagnostics retain the post-schedule normal tails.

B. Extracted Feature Set

Table XII summarizes the extracted features used in the proposed LF/HF/TD pipeline. Each channel is represented using low-frequency spectral features, high-frequency spectral-envelope features, and time-domain statistical features.

C. Severity-Conditioned Attack Detection

Table XIII separates correct bias-bin classification from detecting any attack. The repeated false-alarm entries belong to the same four-class model and do not depend on the actual attack level. Fig. 6 presents the corresponding PCC decision matrices; pooled normal recall can differ slightly from the mean per-repeat recall because normal-window counts vary across holdouts.

D. Hyperparameter Sensitivity

Fig. 10 reports all one-factor perturbations. PCC binary mean balanced accuracy ranges from 72.23% to 78.33%

TABLE XII
EXTRACTED FEATURES PER CHANNEL IN THE PROPOSED LF/HF/TD PIPELINE.

Type	No.	Window / Stride	Features
LF	18	500 / 25	A_1, f_1, ϕ_1, THD, H_2/H_1, H_3/H_1, H_5/H_1, H_7/H_1, $\phi_3-\phi_1$, $\phi_5-\phi_1$, sideband energy/ratio, E_{0-120}, $E_{120-300}$, $E_{300-500}$, df_1/dt, dA_1/dt, $d\mathrm{THD}/dt$
HF	11	250 / 29	RMS_{2-5k}, energy ratio, centroid, bandwidth, flatness, kurtosis, peak frequency, peak-to-band, envelope RMS, envelope kurtosis, envelope crest factor
TD	8	500 / 25	RMS, standard deviation, crest factor, kurtosis, skewness, peak-to-peak, line length, Teager MAV
Total	37	–	LF (18), HF (11), TD (8)

TABLE XIII
EXACT-SEVERITY AND ANY-ATTACK RECALL FROM FOUR-CLASS PREDICTIONS (%).

True bias	Exact recall	Any-attack recall	Normal false alarms
MOTOR stream			
Low	89.58 (±2.14)	95.58 (±4.15)	19.08 (±16.08)
Medium	93.96 (±1.05)	99.71 (±0.04)	19.08 (±16.08)
High	98.76 (±0.27)	99.65 (±0.29)	19.08 (±16.08)
PCC stream			
Low	38.33 (±7.04)	61.39 (±2.64)	37.23 (±3.17)
Medium	50.58 (±7.01)	85.08 (±6.52)	37.23 (±3.17)
High	70.79 (±2.28)	88.09 (±7.22)	37.23 (±3.17)

Mean above (± SD). False alarms refer to the same normal class for all three attack levels. Different metrics are not competing methods, so none is highlighted as best.

among these variants, compared with 80.29% for the selected reference. PCC severity varies from 51.25% to 55.71%, compared with 55.62%. Motor severity is more sensitive to the high learning rate, falling to 77.69% mean balanced accuracy. Therefore, the comparison supports a persistent PCC severity limitation rather than insensitivity to every training setting.

E. Auxiliary Diagnostic Tasks

The motor-type results are included in Table V. For attack-only regression, the target is $100\|\boldsymbol{\alpha}\|_\infty$, in percentage points of nominal current-feedback bias. A scalar output replaces the classification head; training uses Smooth-L1 loss with transition parameter 0.5 and validation mean absolute error (MAE) for checkpoint selection. The severity-selected architecture is retained within each stream and repeat. Motor-side MAE is 0.0196 ± 0.0041 percentage points, with mean $R^2 = 0.961$; PCC MAE is 0.0981 ± 0.0049 percentage points, with mean $R^2 = 0.267$. Fig. 11 shows prediction distributions at every tested bias level for each repeat. These are estimates of injected amplitude on attacked windows, not measured mechanical impact.

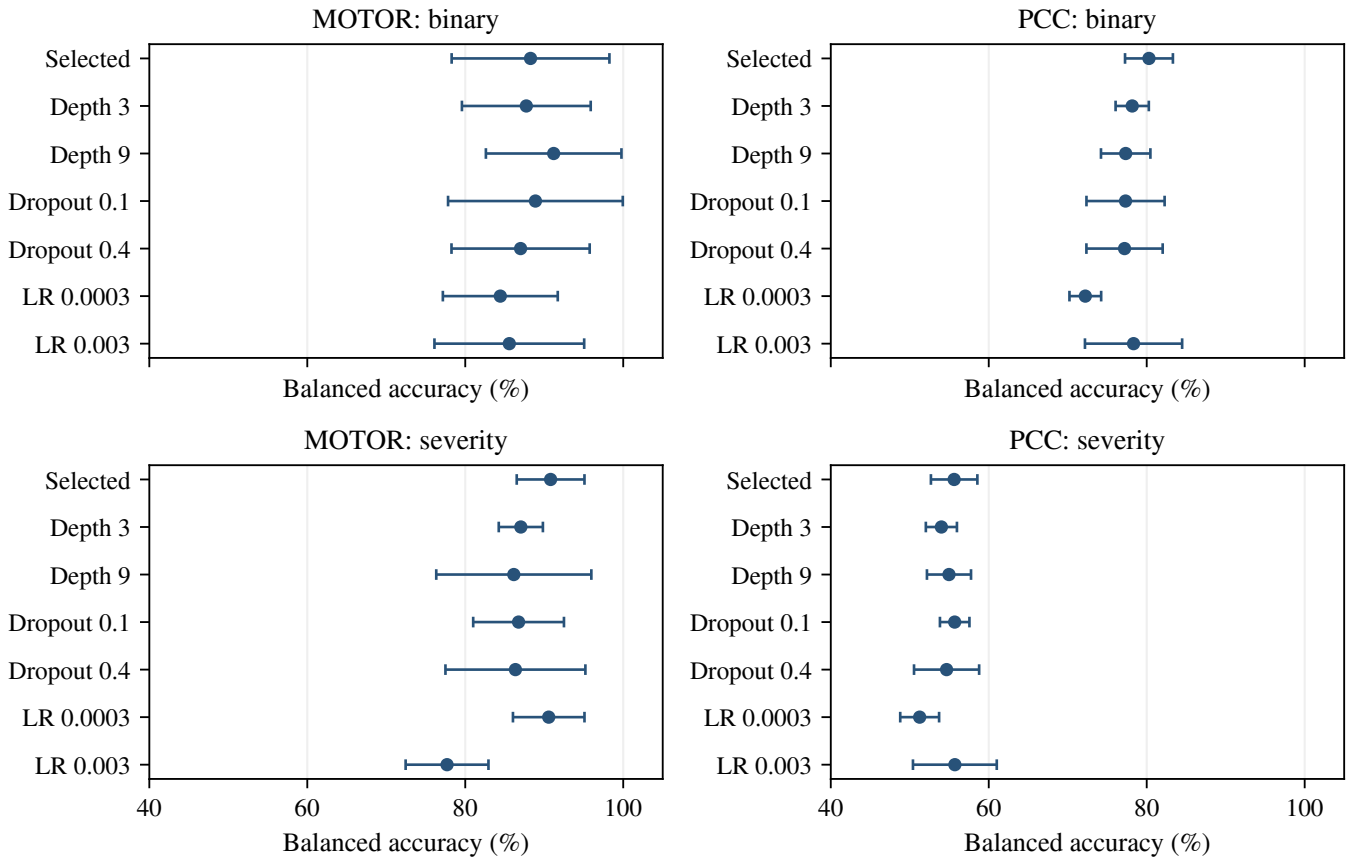


Fig. 10. One-factor sensitivity around the validation-selected full-feature configuration in each repeat. Error bars indicate sample SD across three recording holdouts. Only the named hyperparameter changes; the selected reference can differ across tasks, streams, and repeats.

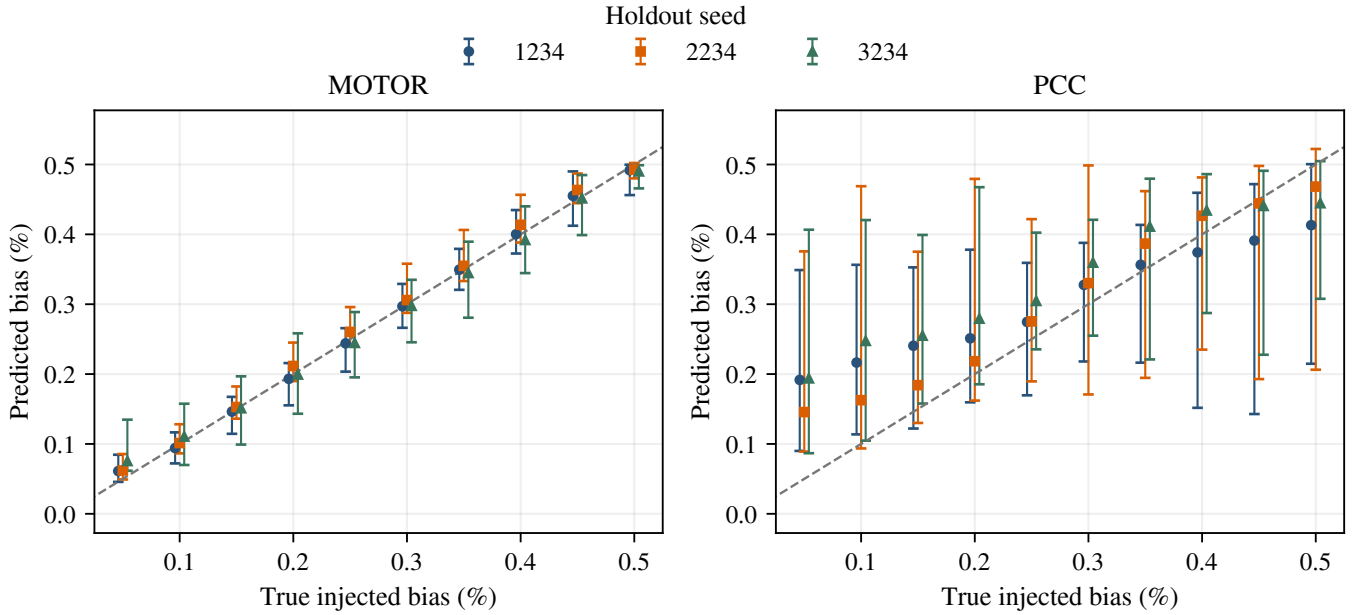


Fig. 11. Attack-only injected-bias regression on held-out recordings. Points and whiskers show the median and 10th–90th percentiles of predictions at each true bias, separately for each holdout seed; slight horizontal offsets separate repeats. The dashed diagonal is ideal agreement. All held-out attack windows are included; repeated appearances across holdouts are not independent recordings.